Experiment No. 5

Program:

DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...

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LINKLIST.C

```
#include <stdio.h>
#include <stdlib.h>
#include <conio.h>
#include <malloc.h>


typedef struct node
{
    int data;
    struct node *next;
} node;
node *createList();
node *Insert_beg(node *head, int x);
node *Delete_beg(node *head);
void PrintList(node *head);


void main()
{
    int choice, insert_option, delete_option, x;
    node *head = NULL;
```

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F1 Help F2 Save F3 Open Alt-F9 Compile F9 Make F10 Menu

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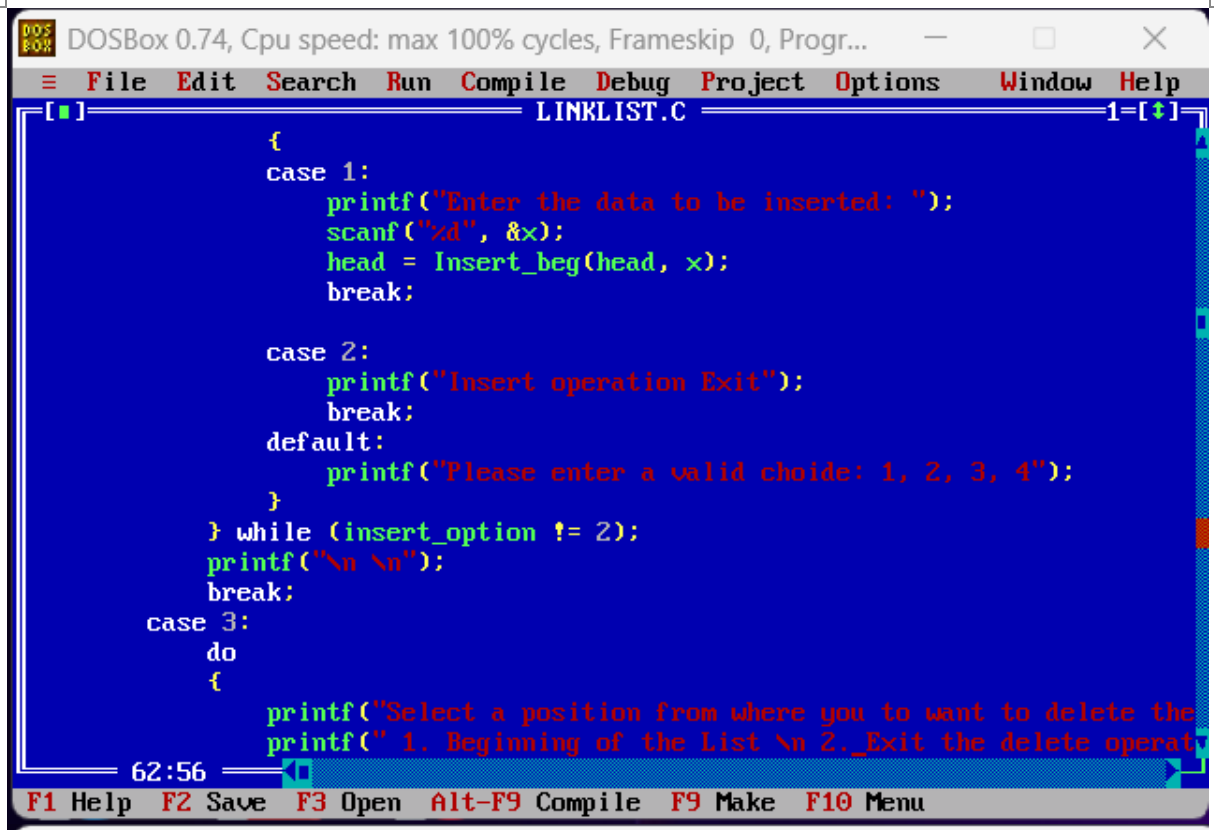
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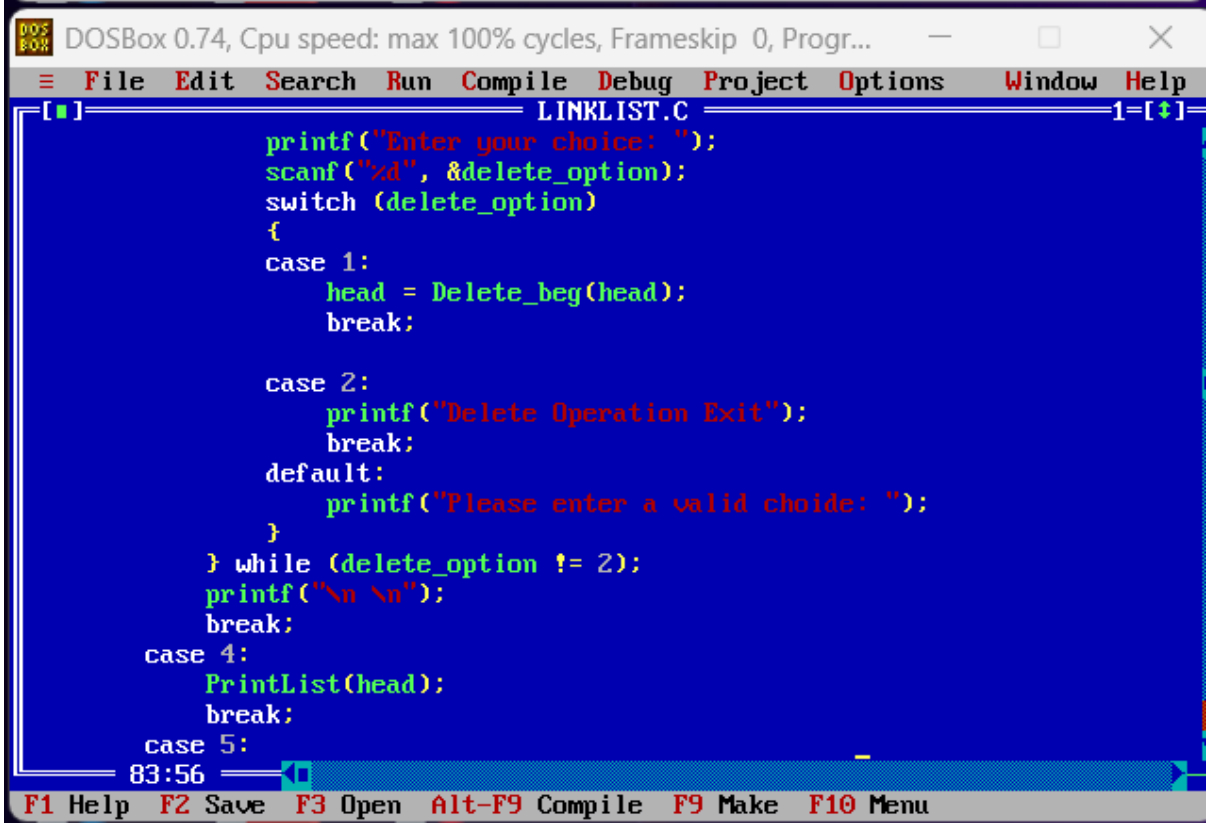
LINKLIST.C

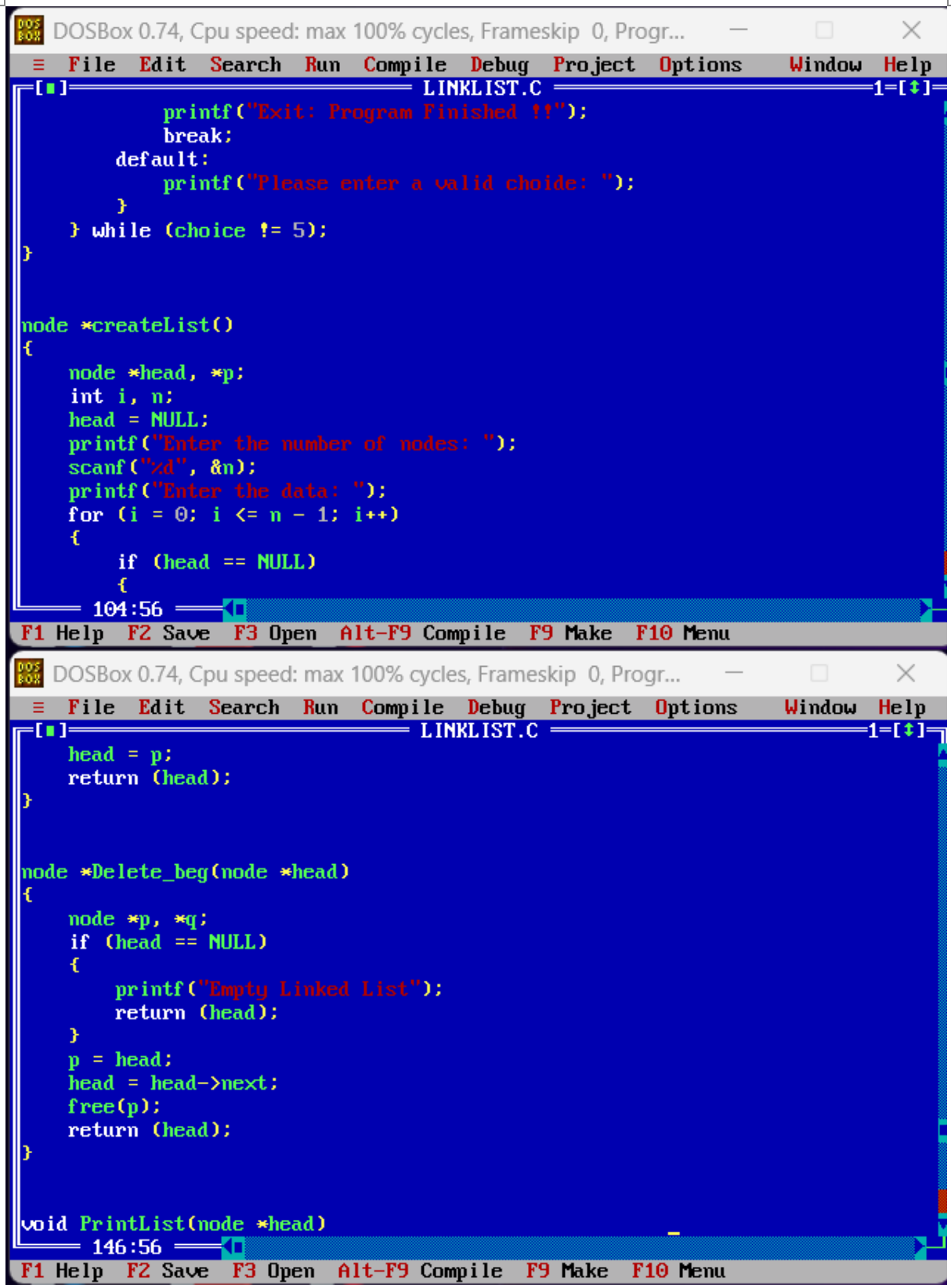
```
    printf("Welcome to the implementation of the singly linked list ! \n");
    do
    {
        printf("Please select an operation to perform from the below list \n")
        printf(" 1. Create a List \n 2. Insert a node \n 3. Delete a node \n 4
        printf("Enter your choice: ");
        scanf("%d", &choice);
        printf("\n \n");
        switch (choice)
        {
        case 1:
            head = createList();
            break;
        case 2:
            do
            {
                printf(" 1. Beginning of the List \n 2. Exit the insert operat
                printf("Enter your choice: ");
                scanf("%d", &insert_option);
                switch (insert_option)
                {
```

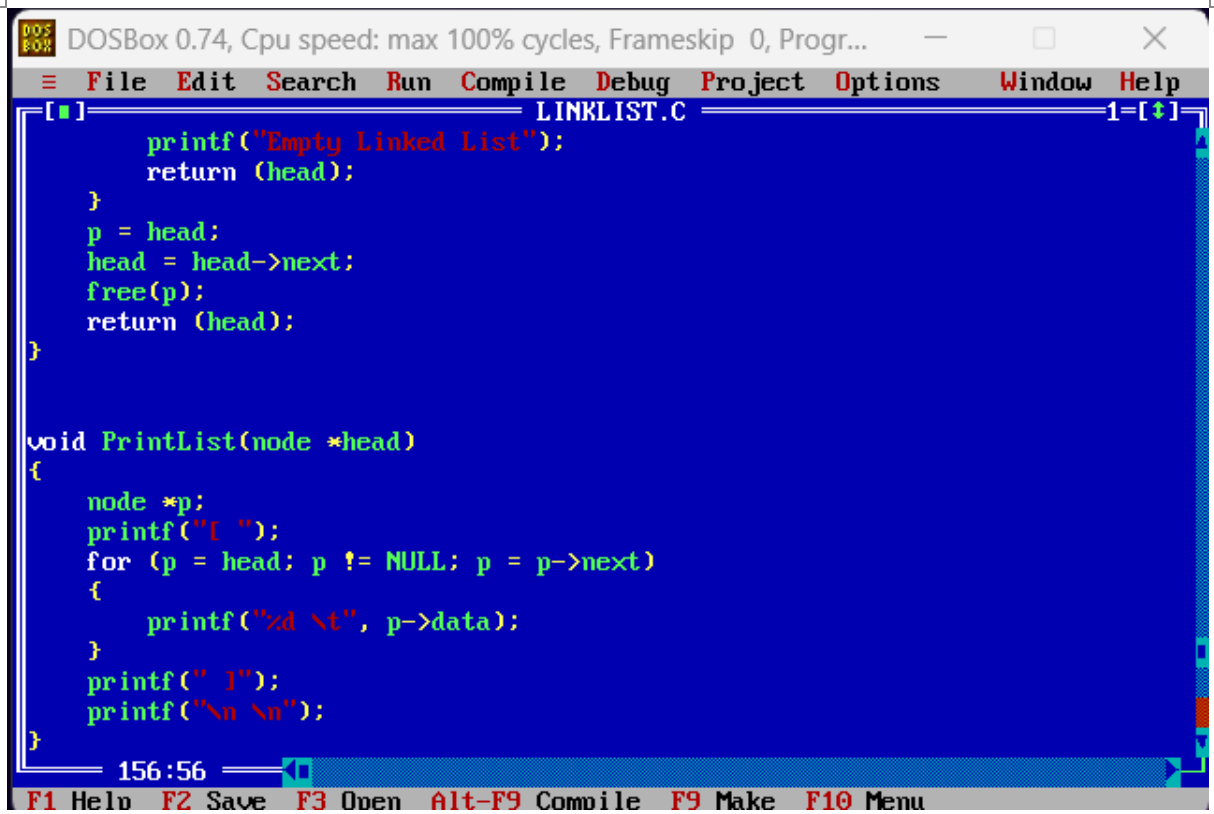
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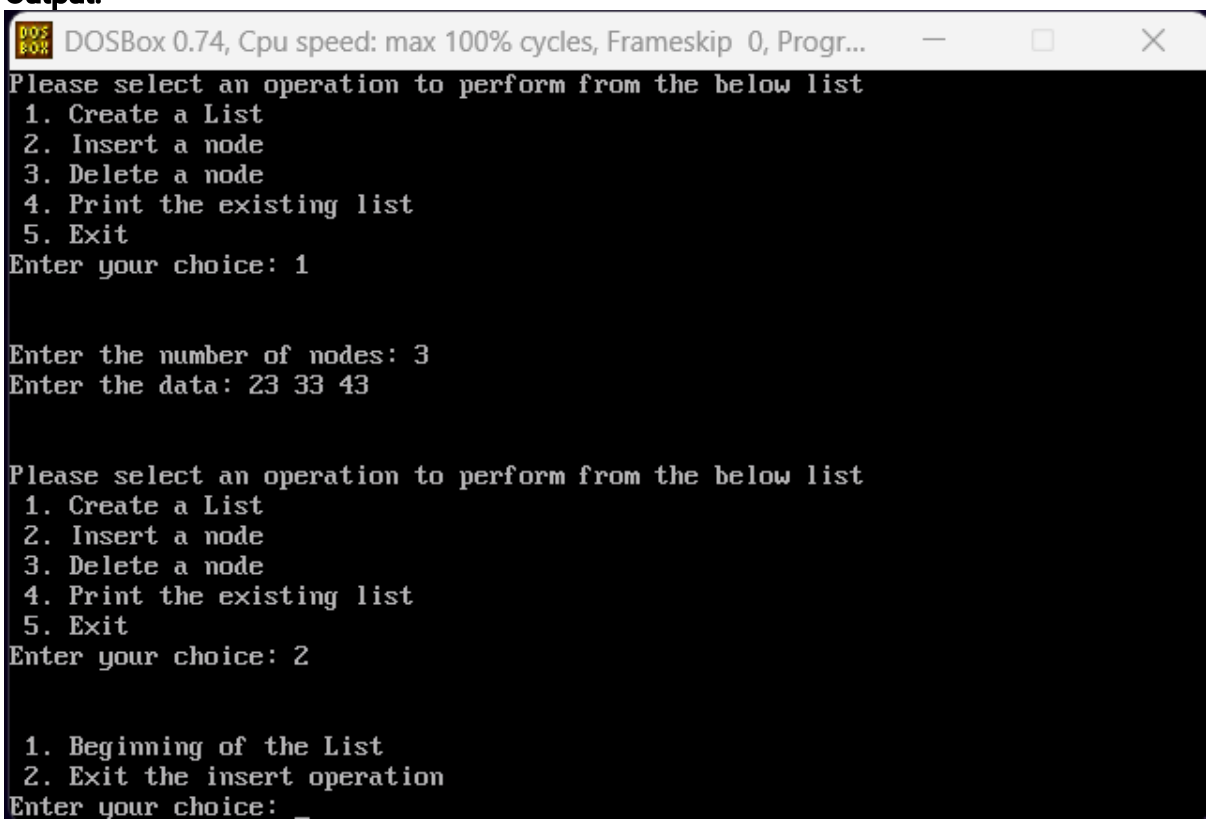

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LINKLIST.C
{
case 1:
printf("Enter the data to be inserted: ");
scanf("%d", &x);
head = Insert_beg(head, x);
break;
case 2:
printf("Insert operation Exit");
break;
default:
printf("Please enter a valid choide: 1, 2, 3, 4");
}
} while (insert_option != 2);
printf("\n \n");
break;
case 3:
do
{
printf("Select a position from where you to want to delete the
printf(" 1. Beginning of the List \n 2._Exit the delete operat
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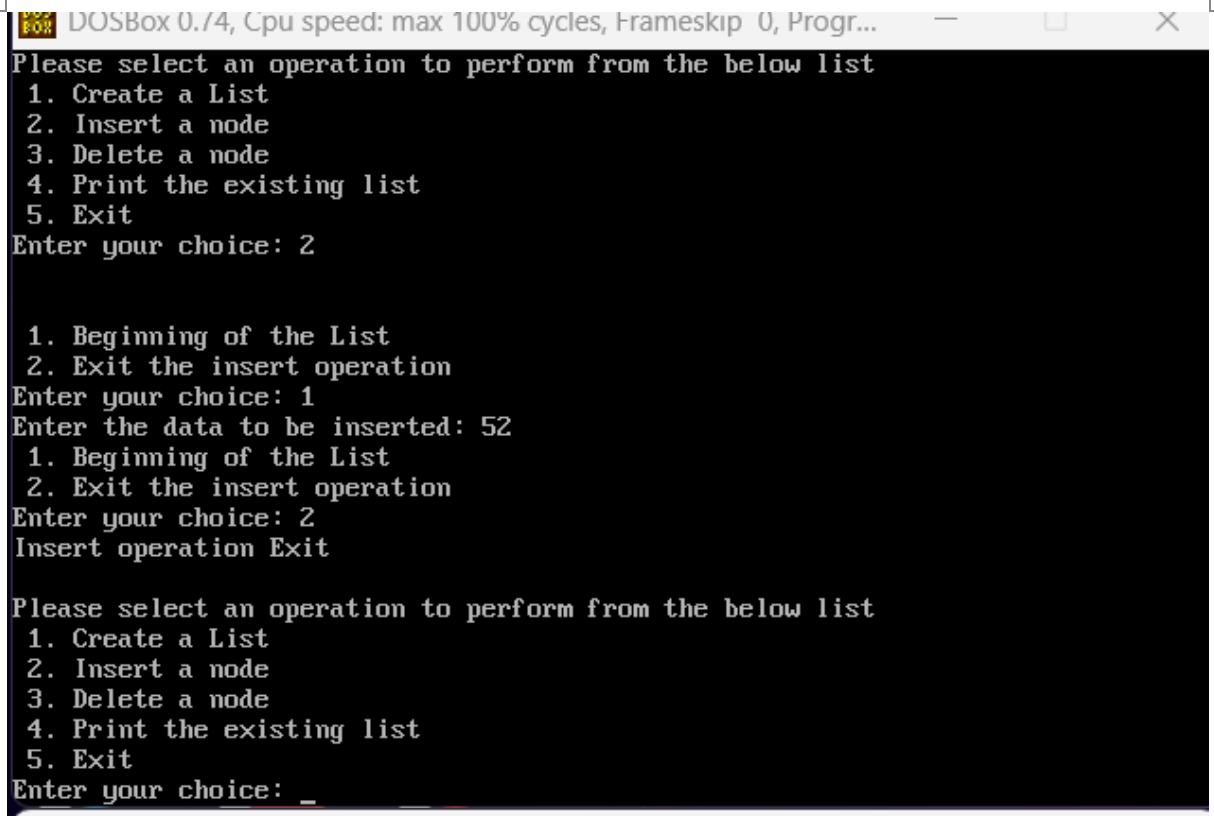

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LINKLIST.C
printf("Enter your choice: ");
scanf("%d", &delete_option);
switch (delete_option)
{
case 1:
head = Delete_beg(head);
break;
case 2:
printf("Delete Operation Exit");
break;
default:
printf("Please enter a valid choide: ");
}
} while (delete_option != 2);
printf("\n \n");
break;
case 4:
PrintList(head);
break;
case 5:
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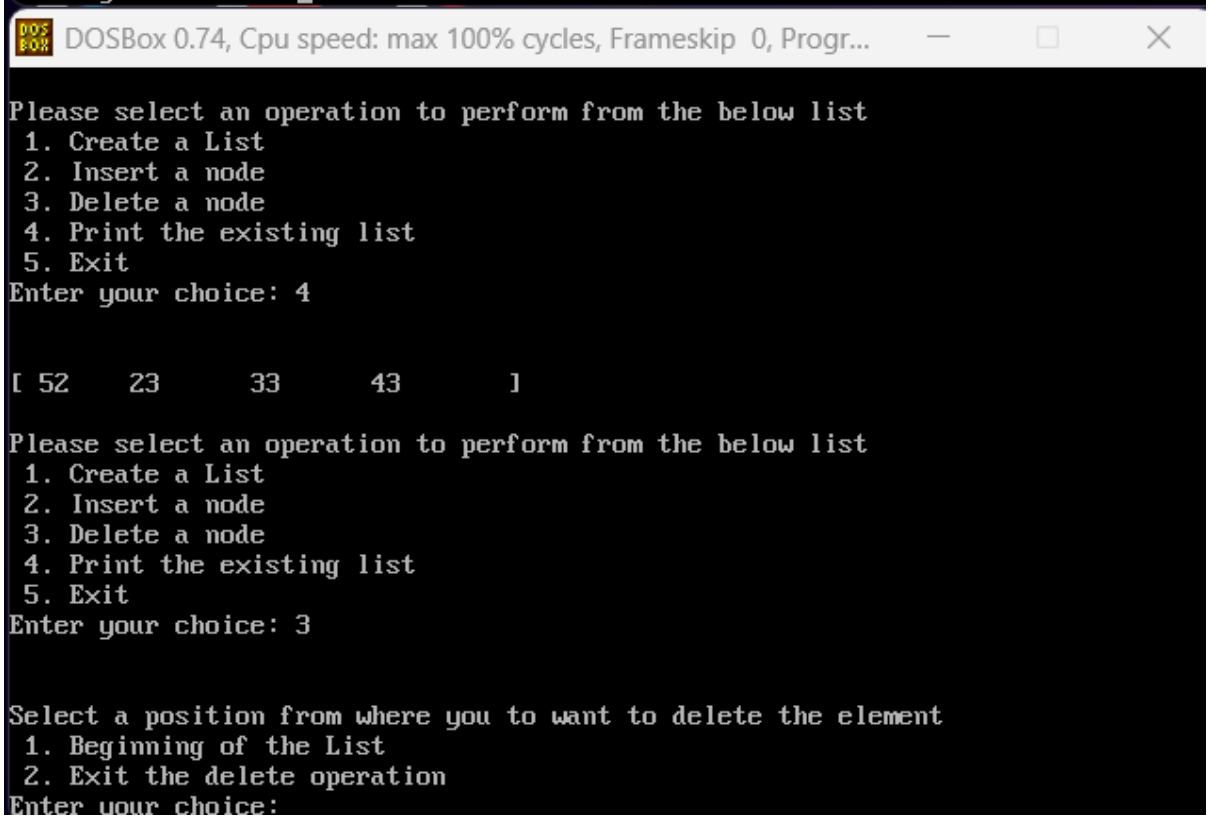

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LINKLIST.C
printf("Exit: Program Finished !!");
break;
default:
printf("Please enter a valid choide: ");
} while (choice != 5);
}
node *createList()
{
node *head, *p;
int i, n;
head = NULL;
printf("Enter the number of nodes: ");
scanf("%d", &n);
printf("Enter the data: ");
for (i = 0; i <= n - 1; i++)
{
if (head == NULL)
{
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DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
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LINKLIST.C
head = p;
return (head);
}
node *Delete_beg(node *head)
{
node *p, *q;
if (head == NULL)
{
printf("Empty Linked List");
return (head);
}
p = head;
head = head->next;
free(p);
return (head);
}
void PrintList(node *head)
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F1 Help F2 Save F3 Open Alt-F9 Compile F9 Make F10 Menu

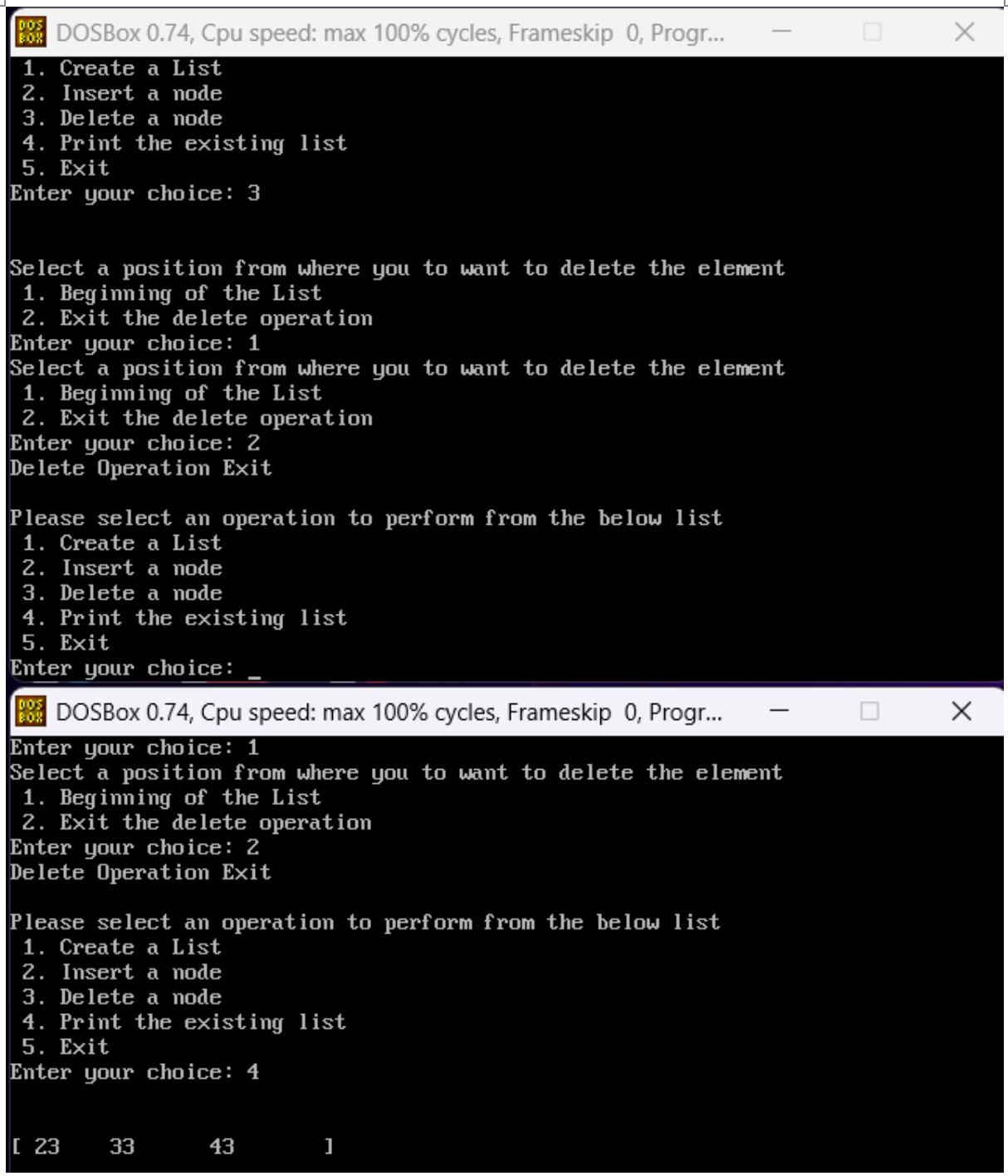

DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
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LINKLIST.C
printf("Empty Linked List");
return (head);
}
p = head;
head = head->next;
free(p);
return (head);
}
void PrintList(node *head)
{
node *p;
printf("[");
for (p = head; p != NULL; p = p->next)
{
printf("%d \t", p->data);
}
printf("]");
printf("\n \n");
}
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F1 Help F2 Save F3 Open Alt-F9 Compile F9 Make F10 Menu

Output:


DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
Please select an operation to perform from the below list
1. Create a List
2. Insert a node
3. Delete a node
4. Print the existing list
5. Exit
Enter your choice: 1
Enter the number of nodes: 3
Enter the data: 23 33 43
Please select an operation to perform from the below list
1. Create a List
2. Insert a node
3. Delete a node
4. Print the existing list
5. Exit
Enter your choice: 2
1. Beginning of the List
2. Exit the insert operation
Enter your choice:


DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
Please select an operation to perform from the below list
1. Create a List
2. Insert a node
3. Delete a node
4. Print the existing list
5. Exit
Enter your choice: 2
1. Beginning of the List
2. Exit the insert operation
Enter your choice: 1
Enter the data to be inserted: 52
1. Beginning of the List
2. Exit the insert operation
Enter your choice: 2
Insert operation Exit
Please select an operation to perform from the below list
1. Create a List
2. Insert a node
3. Delete a node
4. Print the existing list
5. Exit
Enter your choice:


DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
Please select an operation to perform from the below list
1. Create a List
2. Insert a node
3. Delete a node
4. Print the existing list
5. Exit
Enter your choice: 4
[52 23 33 43]
Please select an operation to perform from the below list
1. Create a List
2. Insert a node
3. Delete a node
4. Print the existing list
5. Exit
Enter your choice: 3
Select a position from where you to want to delete the element
1. Beginning of the List
2. Exit the delete operation
Enter your choice:


DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
1. Create a List
2. Insert a node
3. Delete a node
4. Print the existing list
5. Exit
Enter your choice: 3
Select a position from where you to want to delete the element
1. Beginning of the List
2. Exit the delete operation
Enter your choice: 1
Select a position from where you to want to delete the element
1. Beginning of the List
2. Exit the delete operation
Enter your choice: 2
Delete Operation Exit
Please select an operation to perform from the below list
1. Create a List
2. Insert a node
3. Delete a node
4. Print the existing list
5. Exit
Enter your choice: _
DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
Enter your choice: 1
Select a position from where you to want to delete the element
1. Beginning of the List
2. Exit the delete operation
Enter your choice: 2
Delete Operation Exit
Please select an operation to perform from the below list
1. Create a List
2. Insert a node
3. Delete a node
4. Print the existing list
5. Exit
Enter your choice: 4
[23 33 43]